Climate

SWAT+ Documentation

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Purpose

The climate files tell SWAT+ where each spatial object gets its weather. Every object names a weather station in the `wst` column of its connect file (for example `hru.con`). The weather station, a row of `weather-sta.cli`, names a weather generator station and, for each weather variable, either an observed data file or `sim` to generate the variable.

The files below are named on the `climate` line of `file.cio`, in this order:

Position	Default name	Contents
1	weather-sta.cli	Weather stations.
2	weather-wgn.cli	Weather generator statistics.
3	pet.cli	List of observed potential evapotranspiration files.
4	pcp.cli	List of precipitation files (*.pcp).
5	tmp.cli	List of temperature files (*.tmp).
6	slr.cli	List of solar radiation files.
7	hmd.cli	List of relative humidity files.
8	wnd.cli	List of wind speed files.
9	atmodep.cli	Atmospheric deposition of nitrogen.

Two more deposition files, salt_atmo.cli and cs_atmo.cli, have fixed names and are not listed in file.cio.

A position set to null, or naming a file that does not exist, is skipped. A model needs weather-sta.cli and, for any generated variable or missing value, weather-wgn.cli. SWAT+ 62.0.1 reads climate input only from these text files; it has no netCDF climate input.

Common rules

All climate files are free format, with values separated by spaces or tabs. The first line is a title and is skipped; most files also skip a column header line, as described for each file.

Names (stations, files) can be up to 50 characters and are matched with upper and lower case distinguished. A name that contains a slash, a comma or a space must be put in quotes.

SWAT+ looks names up with a binary search, so every list that is searched must be in ascending order by character code (digits before upper-case letters, upper case before lower case). This applies to the station names in weather-sta.cli and weather-wgn.cli, the file names in each observed data list, and the station names in atmodep.cli. A list out of order can make SWAT+ miss names that are present.

Path prefixes

The last lines of file.cio give an optional folder for each type of observed data file: pcp_path, tmp_path, slr_path, hmd_path and wnd_path. When the value is null, the data files are opened from the project folder. Otherwise the value is put in front of each file name with nothing added in between, so it must end with a path separator (for example C:\weather\pcp\), and it can be up to 80 characters. A path containing a forward slash must be put in quotes, because an unquoted slash ends a free-format value. There is no path line for PET files; they are always opened from the project folder.

weather-sta.cli

One line per weather station. Line 1 is a title and line 2 is the column header; both are skipped. Each later line has nine values in the order below.

#	Column	Description
1	name	Weather station name, referenced by the wst column of the connect files.
2	wgn	Name of a weather generator station in weather-wgn.cli.
3	pcp	Name of a precipitation file listed in pcp.cli, or sim to generate precipitation.
4	tmp	Name of a temperature file listed in tmp.cli, or sim to generate maximum and minimum temperature.
5	slr	Name of a solar radiation file listed in slr.cli, or sim.
6	hmd	Name of a relative humidity file listed in hmd.cli, or sim.
7	wnd	Name of a wind speed file listed in wnd.cli, or sim.

#	Column	Description
8	wnd_dir	Read as the name of a PET file listed in pet.cli, or null for none. SWAT+ Editor labels this column wnd_dir; SWAT+ 62.0.1 has no wind direction input.
9	atmo_dep	Name of a deposition station in atmodep.cli, or null.

A tenth value on a line is ignored, so a PET file name placed after atmo_dep is not read.

Unmatched names

Column	Name not found	Message in diagnostics.out
pcp, tmp, slr, hmd, wnd	The variable is generated for the whole run.	<name> file not found (pgage), (tgage), (sgage), (hgage) or (wgage), unless the value is sim
wnd_dir (PET)	No observed PET for the station.	<name> file not found (petgage), unless the value is null; sim gives the message
atmo_dep	See deposition stations.	<name> file not found (atmodep), unless the value is null
wgn	The station uses the first station in weather-wgn.cli.	none

An object whose wst name is not in weather-sta.cli uses the first weather station, also without a message.

The six-day air temperature record used for stream temperature starts each station with the same value, the mean of the January tmp_min_ave and tmp_max_sd of the first station in weather-wgn.cli. It is replaced by the daily mean temperatures of the run within the first six days.

Example

refdata/Ames_sub1/weather-sta.cli in the SWAT+ repository:

```
weather-sta.cli:
name                          wgn              pcp              tmp              slr              hm
wgn_sta1                  wgn_001         ames.pcp         ames.tem              sim              si
```

One station, wgn_sta1, with measured precipitation from ames.pcp and generated solar radiation, humidity and wind. Two names do not match. The generator station in weather-wgn.cli is called wgn001, so wgn_001 is not found and the first (and only) generator station is used without a message. tmp.cli lists ames.tmp, so ames.tem is not found and temperature is generated for the whole run (see *.tem).

weather-wgn.cli

Monthly weather statistics for the weather generator. Each weather station uses its generator station to generate the variables set to sim, to fill missing days in the observed files, and for quantities that depend on location. Line 1 is a title. Each generator station then takes 14 lines:

1. a station line with five values (no header line above it),
2. a column header line, which is skipped,
3. twelve monthly lines, January to December, with 14 values each.

The number of stations is set by the length of the file.

Station line

#	Column	Units	Description
1	name	-	Generator station name, referenced by the wgn column of weather-sta.cli.

#	Column	Units	Description
2	`lat`	decimal degrees	Latitude. Sets day length, maximum possible solar radiation and the day length at which plants go dormant for every weather station that uses this generator station, also when all variables are observed. When SWAT+ estimates plant start dates, a latitude of 0 or less is treated as the Southern Hemisphere.
3	`lon`	decimal degrees	Longitude. Not used by SWAT+.
4	`elev`	m	Elevation. Used for the elevation adjustment of generated precipitation and temperature.
5	`rain_yrs`	years	Number of years of record used to find `pcp_hhr`. A value below 1 is replaced by 10. Called `yrs_pcp` in earlier documentation.

Monthly lines

The column headers below are those SWAT+ Editor writes; older files use the upper-case names in the second column. SWAT+ reads by position only.

#	Column	Older header	Units	Typical range	Description
1	`tmp_max_ave`	`TMPMX`	°C	-30 to 50	Mean daily maximum air temperature for the month.
2	`tmp_min_ave`	`TMPMN`	°C	-40 to 40	Mean daily minimum air temperature for the month.
3	`tmp_max_sd`	`TMPSTDMX`	°C	0.1 to 100	Standard deviation of daily maximum air temperature in the month.
4	`tmp_min_sd`	`TMPSTDMN`	°C	0.1 to 30	Standard deviation of daily minimum air temperature in the month.
5	`pcp_ave`	`PCPMM`	mm	0 to 600	Mean total precipitation for the month.
6	`pcp_sd`	`PCPSTD`	mm/day	0.1 to 50	Standard deviation of daily precipitation in the month, with dry days included.
7	`pcp_skew`	`PCPSKW`	-	0.2 to 20	Skew coefficient of daily precipitation in the month, with dry days included. Values below 0.2 are replaced by 0.2.
8	`wet_dry`	`PR_WD`	fraction	0 to 0.95	Probability of a wet day following a dry day in the month.
9	`wet_wet`	`PR_WW`	fraction	0 to 0.95	Probability of a wet day following a wet day in the month.
10	`pcp_days`	`PCPD`	days	0 to 31	Mean number of days with precipitation in the month.
11	`pcp_hhr`	`RAINHMX`	mm	0 to 125	Largest 0.5-hour rainfall in the month over the `rain_yrs` years of record.
12	`slr_ave`	`SOLARAV`	MJ/m²/day	0 to 40	Mean daily solar radiation for the month.
13	`dew_ave`	`DEWPT`	°C or fraction	-50 to 25	Mean daily dew point temperature for the month, or mean relative humidity as a fraction.
14	`wnd_ave`	`WINDAV`	m/s	0 to 100	Mean daily wind speed for the month.

The statistics count a day as wet when it has more than 0 mm of precipitation. During the run, the generator counts a day as wet, for the next day's wet-day probability, when it has at least 0.1 mm.

Checks and derived values

SWAT+ checks and completes the monthly values for each generator station when it reads the file.

Wet-day probabilities and wet days. If `wet_wet` is not greater than `wet_dry`, or `wet_dry` is 0 or less, the two probabilities are estimated from `pcp_days` (at least 0.1 day):

$$P(W/D) = 0.75\,\frac{d_{wet}}{N}, \qquad P(W/W) = 0.25 + P(W/D)$$

where d_{wet} is `pcp_days` and N the number of days in the month. Otherwise `pcp_days` is not used as given and is recalculated from the probabilities:

$$d_{wet} = \frac{N\,P(W/D)}{1 - P(W/W) + P(W/D)}$$

Mean precipitation per wet day. The generator draws daily amounts around $\mu = \text{pcp_ave}/d_{wet}$. `pcp_ave` must therefore be the mean monthly total. A mean daily value in this column makes generated precipitation far too small.

Dew point or humidity. If all twelve `dew_ave` values are between 0 and 1, they are taken as relative humidity (fraction). If any value is outside that range, all twelve are taken as dew point temperature in °C.

Half-hour rainfall. `pcp_hhr` is smoothed over three months (the month and its two neighbours, December and January wrapping round) and turned into the mean fraction of daily rain that falls in the wettest half hour:

$$\alpha_{0.5} = \text{adj_pkr}\left[1 - \exp\left(\frac{\bar{R}_{0.5}}{\mu\,\ln\left(\dfrac{0.5}{\text{rain_yrs}\;d_{wet}}\right)}\right)\right]$$

limited to 0.1 to 0.95, where $\bar{R}_{0.5}$ is the smoothed `pcp_hhr` and `adj_pkr` is the peak rate adjustment factor (`adj_pkrt` in the `parameters.bsn` header). Each day's fraction is drawn at random around this monthly value and is used for peak runoff rates, erosion and the generated distribution of rain within the day.

The generator equations themselves are described in Climate and weather.

Example

Start of `refdata/Ames_sub1/weather-wgn.cli` in the SWAT+ repository:

```
weather-wgn.cli: Ames
wgn001       37.48     -100.83    887.00     10
   TMPMX   TMPMN  TMPSTDMX TMPTDMN  PCPMM  PCPSTD  PCPSKW   PR_WD   PR_WW   PCPD  RAINHMX SOLARAV  DEWPT
    7.61   -8.13     8.06    5.99     7.4     3.3    0.19    0.07    0.25    2.65      5.1   10.39  -8.14
   11.07   -5.53     8.06    5.34    10.2     5.1    2.67    0.08    0.27    2.86      1.8   13.41  -5.58
   15.22   -1.87     7.99    5.22    28.1     8.9    1.64    0.11    0.34    4.43      7.6   17.85  -4.71
...
```

One generator station, `wgn001`, at 37.48° N and 887 m, with 10 years of half-hour rainfall record; ten more monthly lines follow. The January skew of 0.19 is raised to 0.2. The `DEWPT` values go below 0, so the column is read as dew point temperature.

Observed data lists

`pcp.cli`, `tmp.cli`, `slr.cli`, `hmd.cli`, `wnd.cli` and `pet.cli` all have the same layout: a title line, a header line (SWAT+ Editor writes `filename`), and one data file name per line. Each name is both the file to open and the name that the matching column of `weather-sta.cli` refers to. Every file listed is read, whether or not a station uses it.

List	`weather-sta.cli` column	Records in each data file	Units
`pcp.cli`	`pcp`	year, day, precipitation (or six values for sub-daily data)	mm (`*.pcp`)
`tmp.cli`	`tmp`	year, day, maximum, minimum temperature	°C (`*.tmp`)
`slr.cli`	`slr`	year, day, solar radiation	MJ/m²/day
`hmd.cli`	`hmd`	year, day, relative humidity	fraction, 0 to 1
`wnd.cli`	`wnd`	year, day, wind speed	m/s, at 10 m height
`pet.cli`	`wnd_dir`	year, day, potential evapotranspiration	mm

Example, `refdata/Osu_1hru/pcp.cli`:

```
pcp.cli: Precipitation file names - file written by SWAT+ editor 2023-03-22 04:22:29.020522
filename
Imsilpcp.pcp
```

Solar radiation, humidity, wind and PET files

These data files have the layout of a daily precipitation file: a title line, a header line (`nbyr tstep lat lon elev`), a gauge line with those five values, and one record per day with year, day of year (called `jday` in earlier documentation) and the value. Only `nbyr` is used from the gauge line. The records follow the same rules for order, storage, values of -97 or less, and days outside the record as daily precipitation. A missing solar radiation, humidity or wind value is generated from the weather generator station.

Relative humidity is limited to 0.01 to 0.99 every day after it is read or generated, so a file in percent gives 0.99 on every day. Solar radiation below 0 is set to 0.

Observed PET

A PET file is read when it is listed in `pet.cli`, and a station uses it when its `wnd_dir` column names it. With `pet` = 3 in `codes.bsn`, HRUs take their potential evapotranspiration from the station. In 62.0.1 the station value is replaced by a Hargreaves estimate (at least 0.01 mm) later in the same daily step, for every station, so the values in the PET files do not reach the HRUs and `pet` = 3 gives Hargreaves PET. A missing PET value is also replaced by a Hargreaves estimate. Channel and reservoir evaporation and aquifer revap use the same station value under every `pet` setting.

atmodep.cli

Atmospheric deposition of ammonium and nitrate, as average annual, monthly or yearly values for one or more stations. Earlier documentation calls this file `atmo.cli`. The deposition values are in this file; it does not point to data files. The `atmo_dep` column of `codes.bsn` is not used; the time step comes from this file.

Line 1 is a title and line 2 a column header; both are skipped. Line 3 is the control line:

#	Column	Description
1	`num_sta`	Number of deposition stations in the file.
2	`timestep`	`aa` average annual, `mo` monthly, `yr` yearly.
3	`mo_init`	First month of the series (1 to 12). Used with `mo`.
4	`yr_init`	First year of the series. Used with `mo` and `yr`.
5	`num_aa`	Number of months (`mo`) or years (`yr`) in each series. Not used with `aa`.

#	Column	Description

Each station then takes five lines:

Line	Contents	Units
1	Station name, referenced by the `atmo_dep` column of `weather-sta.cli`.	-
2	Ammonium concentration in rain.	mg N/L
3	Nitrate concentration in rain.	mg N/L
4	Ammonium dry deposition.	kg N/ha per year (`aa`, `yr`) or per month (`mo`)
5	Nitrate dry deposition.	kg N/ha per year (`aa`, `yr`) or per month (`mo`)

With `aa` each of lines 2 to 5 has one value. With `mo` or `yr` each has `num_aa` values, one per month or year from `mo_init`/`yr_init` on.

Daily deposition

Every day, each HRU receives in its top soil layer, for ammonium and nitrate separately,

$$N_{wet} = 0.01\, C\, P, \qquad N_{dry} = \frac{D}{365}\ \ (\texttt{aa}, \texttt{yr}), \qquad N_{dry} = \frac{D}{N_{mo}}\ \ (\texttt{mo})$$

where N_{wet} and N_{dry} are in kg N/ha, C is the concentration in rain (mg/L), P the HRU precipitation for the day (mm), D the dry deposition value and N_{mo} the number of days in the month.

Monthly and yearly series

SWAT+ starts the series at the entry for the first month or year of the simulation. If the series starts later than the simulation, deposition begins when its first month or year is reached. Before the series starts and after its last entry, HRUs receive no deposition. The entry used moves on at the last day of each month (`mo`) or year (`yr`), so the last day of a month already uses the next month's values.

When the series starts before the simulation and `mo_init` is not 1, the first entry used is `mo_init` - 1 months later than the simulation start month, and the whole series is shifted by that amount.

Deposition stations

A weather station with `atmo_dep` set to `null`, or to a name not in the file, uses a built-in station. With `aa` this station gives 1.0 mg/L ammonium and 0.2 mg/L nitrate in rain and no dry deposition. With `mo` or `yr` the built-in station has no series, and its values are undefined; give every weather station a deposition station in that case.

Example

Average annual deposition for one station:

```
atmodep.cli
num_sta  timestep  mo_init  yr_init  num_aa
      1        aa        0        0       0
station_001
1.0
0.2
0.0
0.0
```

Salt and constituent deposition

Wet and dry deposition of salt ions and of other constituents, used with the salt and constituent inputs. Both files have fixed names and are read from the project folder only when `atmodep.cli` is present and the corresponding ions or constituents are defined in `constituents.cs`. They have no control line: the number of stations, the time step and the series length come from `atmodep.cli`. Stations are matched by position, so they must be in the same order as in `atmodep.cli`.

salt_atmo.cli

The first six lines are skipped. For each station:

- a line with the station name,
- one line per salt ion, in the order of `constituents.cs`, with the ion label followed by the concentration in rain (mg/L), one value for `aa` or `num_aa` values for `mo` and `yr`,
- one line per salt ion with the ion label followed by the dry deposition, in the same units as nitrogen dry deposition.

cs_atmo.cli

The first three lines are skipped. For each station:

- a station line, which is read as a number, so it must contain a number such as the station's position (a text name stops the run with a read error),
- one line per constituent with the concentration in rain (mg/L), without a label,
- one line per constituent with the dry deposition, without a label.

For both files, the daily amounts are calculated as for nitrogen and added to the top soil layer.

Related files

- `file.cio`: names the climate files and the data path prefixes.
- `hru.con` and the other connect files: assign a weather station to each object in the `wst` column.
- `*.pcp`, `*.tmp`, `*.tem`: observed precipitation and temperature.
- `time.sim`: simulation period and number of steps per day.
- `codes.bsn`: PET method and `lapse` switch.
- `parameters.bsn`: lapse rates and `adj_pkrt`.
- Salt inputs and constituent inputs: ion and constituent lists for the deposition files.
- Climate and weather: weather generator and PET methods.

Sources

- File name slots: `type input_cli` and the `in_path_*` types in `input_file_module.f90`, read in `readcio_read.f90`
- Type definitions: `weather_station`, `weather_codes_station_char`, `weather_generator_db`, `climate_measured_data`, `atmospheric_deposition`, `atmospheric_deposition_control` in `climate_module.f90`
- `weather-sta.cli`: `cli_staread.f90`; name lookup `search.f90`; station lookup from connect files `hyd_read_connect.f90`
- `weather-wgn.cli`: `cli_wgnread.f90`, `cli_initwgn.f90`, `cli_clgen.f90`, `cli_tgen.f90`
- Observed data lists: `cli_pmeas.f90`, `cli_tmeas.f90`, `cli_smeas.f90`, `cli_hmeas.f90`, `cli_wmeas.f90`, `cli_petmeas.f90`
- Daily climate step, missing values, PET recalculation and humidity limits: `climate_control.f90`; HRU PET `et_pot.f90`
- `atmodep.cli`: `cli_read_atmodep.f90`, `cli_atmodep_time_control.f90`, `nut_nrain.f90`
- `salt_atmo.cli` and `cs_atmo.cli`: `cli_read_atmodep_salt.f90`, `cli_read_atmodep_cs.f90`, `salt_rain.f90`, `cs_rain.f90`
- `atmo_dep` code in `codes.bsn`: `bsn_cc%atmo` in `basin_module.f90`

File	Column	Fortran variable
weather-sta.cli	name	wst(i)%name
weather-sta.cli	wgn	wst(i)%wco_c%wgn
weather-sta.cli	pcp	wst(i)%wco_c%pgage
weather-sta.cli	tmp	wst(i)%wco_c%tgage
weather-sta.cli	slr	wst(i)%wco_c%sgage
weather-sta.cli	hmd	wst(i)%wco_c%hgage
weather-sta.cli	wnd	wst(i)%wco_c%wgage
weather-sta.cli	wnd_dir	wst(i)%wco_c%petgage
weather-sta.cli	atmo_dep	wst(i)%wco_c%atmodep
weather-wgn.cli	name	wgn_n(i)
weather-wgn.cli	rain_yrs	wgn(i)%rain_yrs
weather-wgn.cli	tmp_max_ave, tmp_min_ave	wgn(i)%tmpmx, wgn(i)%tmpmn
weather-wgn.cli	tmp_max_sd, tmp_min_sd	wgn(i)%tmpstdmx, wgn(i)%tmpstdmn
weather-wgn.cli	pcp_ave, pcp_sd, pcp_skew	wgn(i)%pcpmm, wgn(i)%pcpstd, wgn(i)%pcpskw
weather-wgn.cli	wet_dry, wet_wet, pcp_days	wgn(i)%pr_wd, wgn(i)%pr_ww, wgn(i)%pcpd
weather-wgn.cli	pcp_hhr	wgn(i)%rainhmx
weather-wgn.cli	slr_ave, dew_ave, wnd_ave	wgn(i)%solarav, wgn(i)%dewpt, wgn(i)%windav
observed lists	file name	pcp_n, tmp_n, slr_n, hmd_n, wnd_n, petm_n and %filename
atmodep.cli	control line	atmodep_cont%num_sta, %timestep, %mo_init, %yr_init, %num
atmodep.cli	aa lines 2 to 5	atmodep(i)%nh4_rf, %no3_rf, %nh4_dry, %no3_dry
atmodep.cli	mo lines 2 to 5	atmodep(i)%nh4_rfmo, %no3_rfmo, %nh4_drymo, %no3_drymo
atmodep.cli	yr lines 2 to 5	atmodep(i)%nh4_rfyr, %no3_rfyr, %nh4_dryyr, %no3_dryyr

Purpose

A precipitation data file contains the measured precipitation of one gauge, as daily totals or as values for each sub-daily time step. Each file is named on one line of pcp.cli, and a weather station uses it when the pcp column of weather-sta.cli gives the same name. Days without a usable value are filled by the weather generator station named in the wgn column of the same weather station.

SWAT+ does not check the file extension. .pcp is the usual one. If pcp_path in file.cio is not null, its value is put in front of every file name in pcp.cli (see path prefixes).

Format

Free format, whitespace-separated. Line 1 is a title and line 2 is a column header; both are skipped. Line 3 describes the gauge and every later line is one record.

Gauge line

SWAT+ Editor writes the header nbyr tstep lat lon elev above this line.

#	Column	Units	Description
1	nbyr	years	Number of years of record. SWAT+ uses it to size the store for the data, so it must be at least the number of calendar years the record spans.
2	tstep	values per day	0 for daily data. For sub-daily data, the number of values per day, which must equal step in time.sim (24 for hourly data).
3	lat	decimal degrees	Latitude of the gauge. Not used by SWAT+.
4	lon	decimal degrees	Longitude of the gauge. Not used by SWAT+.
5	elev	m	Elevation of the gauge. Used only for the elevation adjustment.

Daily records

With tstep = 0, each record has three values:

#	Column	Units	Description
1	year	-	Calendar year.
2	day	-	Day of the year, 1 to 366. Called `jday` in earlier documentation.
3	pcp	mm	Precipitation for the day. A value of -97 or less marks the day as missing.

Sub-daily records

With `tstep` greater than 0, each record has six values:

#	Column	Units	Description
1	year	-	Calendar year.
2	day	-	Day of the year, 1 to 366. Called `jday` in earlier documentation.
3	month	-	Month, called `mon` in earlier documentation. Not used by SWAT+.
4	day of month	-	Not used by SWAT+.
5	step	-	Time step within the day, called `hr` in earlier documentation. Not used by SWAT+.
6	pcp	mm	Precipitation for the time step.

SWAT+ places the values of each day in the order they appear, so every day needs `step` records in time order. The month, day of month and step columns do not position the value.

Record order

Records must be continuous and in date order. SWAT+ counts years by watching for day 365 or 366 followed by a record from a new year, so a gap at the end of a year shifts every later year of the record. A file that starts or ends part way through a year is fine.

Reading and storage

Records before the start of the simulation are skipped when the file is read. Storage starts with the first record whose year is at or after `yrc_start` and whose day is at or after `day_start` in `time.sim`.

When the record starts in a year after the simulation start year and `day_start` is later than the first day in the file, the records of that first year before `day_start` are also skipped. Those days then read as 0 mm instead of being filled by the weather generator.

Missing days

In a daily run, SWAT+ fills a day from the weather generator when

- the value is -97 or less (SWAT+ Editor and most tools write -99), or
- the day is before the first record or after the last record of the file.

Generated precipitation uses the monthly statistics of the station's `weather-wgn.cli` entry and the wet or dry state of the previous day. The method is described in Climate and weather.

Sub-daily runs

A sub-daily file is read when `tstep` is greater than 0. `step` in `time.sim` sets the number of steps per day of the run, and SWAT+ stores that many values per day. If the two numbers differ, the values are assigned to the wrong steps: with `step` = 0 or 1 and an hourly file, only the last hour of each day is kept.

Missing and out-of-range values are handled differently from daily data in 62.0.1:

- When a step has a value of -97 or less, SWAT+ calls the generator, but the day's total is still the sum of the file values for that day, missing-value flags included.
- Days outside the record are not generated. Inside the stored years they read as 0 mm.
- Values are looked up by simulation year counted from the simulation start, so the record must start on or before the simulation start date.
- From the second simulation year on, the values used for 1 January come from 1 January of the previous simulation year.

Sub-daily files should therefore be complete, without missing values, and cover the whole simulation period.

Elevation adjustment

When `lapse` = 1 in `codes.bsn`, the precipitation of each HRU is changed for the difference between the HRU elevation and the gauge elevation:

$$P_{hru} = \max\left(0,\ P + \text{plaps}\,\frac{z_{hru} - z_{gauge}}{1000}\right)$$

where P is the station precipitation (mm), `plaps` is the precipitation lapse rate in `parameters.bsn` (mm per km, default 0), z_{hru} is the `elev` column of `hru.con` and z_{gauge} is `elev` from this file (m). When the station's `pcp` column is `sim`, the elevation of the weather generator station is used for z_{gauge}.

The adjustment is added every day, dry days included, so a positive `plaps` gives HRUs above the gauge a small amount of precipitation on every day. The adjustment values are calculated during channel and reservoir routing, so in a model without channels or reservoirs they remain zero.

Calibration

The calibration parameter `precip` (object type `cli`) changes the stored daily values of listed precipitation records over a window of days and years. See Climate changes.

Example

First lines of `refdata/Ames_sub1/ames.pcp` in the SWAT+ repository, and part of 2012:

```
AME.pcp
 NBYR    TSTEP      LAT     LONG     ELEV
   121        0    42.04   -93.89   316
1900 1 -99
1900 2 -99
1900 3 -99
...
2012 55 12.4
2012 56 0
2012 57 -99
2012 58 0
```

A daily record (`tstep` = 0) of 121 years, 1900 to 2020, at 316 m. The Ames run starts in 1975, so the records from 1900 to 1974 are skipped when the file is read. Of the days from 1975 on, 31 are marked missing (`-99`), such as day 57 of 2012, and the weather generator fills them during the run.

Related files

- `pcp.cli`: lists the precipitation files.
- `weather-sta.cli`: assigns a precipitation file to each weather station in its `pcp` column.
- `weather-wgn.cli`: weather generator statistics that fill missing days.

- time.sim: simulation period and number of steps per day.
- file.cio: names pcp.cli and the pcp_path prefix.
- codes.bsn and parameters.bsn: lapse switch and plaps rate.

Sources

- Reader: cli_pmeas.f90
- Type definition: type climate_measured_data in climate_module.f90
- Gauge name lookup from weather-sta.cli: cli_staread.f90, search.f90
- Daily use, missing values and range check: cli_precip_control.f90, cli_bounds_check.f90, called from time_control.f90
- Weather generator: cli_pgen.f90, cli_pgenhr.f90
- Elevation adjustment: cli_lapse.f90, called from sd_channel_control3.f90 and res_control.f90, applied in hru_control.f90
- Time step: time_read.f90
- Path prefix: in_path_pcp in input_file_module.f90, read in readcio_read.f90
- Calibration change precip: cal_conditions.f90

Column	Fortran variable
nbyr	pcp(i)%nbyr
tstep	pcp(i)%tstep
lat	pcp(i)%lat
lon	pcp(i)%long
elev	pcp(i)%elev
daily pcp	pcp(i)%ts(day, iyr)
sub-daily pcp	pcp(i)%tss(step, day, iyr)

Purpose

A temperature data file contains the measured daily maximum and minimum air temperature of one gauge. Each file is named on one line of tmp.cli, and a weather station uses it when the tmp column of weather-sta.cli gives the same name. Days without usable values are filled by the weather generator station named in the wgn column of the same weather station.

SWAT+ does not check the file extension. SWAT+ Editor writes .tmp; some older data sets use .tem. If tmp_path in file.cio is not null, its value is put in front of every file name in tmp.cli (see path prefixes).

Temperature is always read as daily data, also in sub-daily runs.

Format

Free format, whitespace-separated. Line 1 is a title and line 2 is a column header; both are skipped. Line 3 describes the gauge and every later line is one daily record.

Gauge line

SWAT+ Editor writes the header nbyr tstep lat lon elev above this line.

#	Column	Units	Description
1	nbyr	years	Number of years of record. SWAT+ uses it to size the store for the data, so it must be at least the number of calendar years the record spans.
2	tstep	-	Time step of the data. Not used by SWAT+; give 0.
3	lat	decimal degrees	Latitude of the gauge. Not used by SWAT+.
4	lon	decimal degrees	Longitude of the gauge. Not used by SWAT+.
5	elev	m	Elevation of the gauge. Used only for the elevation adjustment.

Daily records

#	Column	Units	Description
1	year	-	Calendar year.
2	day	-	Day of the year, 1 to 366. Called `jday` in earlier documentation.
3	tmax	°C	Maximum air temperature for the day, called `tmpmax` in earlier documentation. A value of -97 or less marks it as missing.
4	tmin	°C	Minimum air temperature for the day, called `tmpmin` in earlier documentation. A value of -97 or less marks it as missing.

Records must be continuous and in date order, as for precipitation: SWAT+ counts years by watching for day 365 or 366 followed by a record from a new year. Records before the simulation start are skipped when the file is read, by the rule described for precipitation files.

Missing days

SWAT+ generates both temperatures for a day when

- `tmax` or `tmin` is -97 or less (SWAT+ Editor and most tools write -99), or
- the day is before the first record or after the last record of the file.

When only one of the two values is missing, the measured value of the other is replaced by a generated one as well, so the pair always comes from one source. The generated maximum is lower on wet days than on dry days, using the day's precipitation from the same weather station. The method is described in Climate and weather.

The daily mean air temperature used by snow, plant growth and other processes is

$$T_{av} = \frac{T_{max} + T_{min}}{2}$$

Elevation adjustment

When `lapse` = 1 in `codes.bsn`, the temperatures of each HRU are changed for the difference between the gauge elevation and the HRU elevation:

$$T_{hru} = T + \text{tlaps}\,\frac{z_{gauge} - z_{hru}}{1000}$$

applied to the maximum, minimum and mean temperature. `tlaps` is the temperature lapse rate in `parameters.bsn` (°C per km, default 6.5), z_{hru} is the `elev` column of `hru.con` and z_{gauge} is `elev` from this file (m). When the station's `tmp` column is `sim`, the elevation of the weather generator station is used for z_{gauge}. As for precipitation, the adjustment values are calculated during channel and reservoir routing, so in a model without channels or reservoirs they remain zero.

Calibration

The calibration parameter `temp` (object type `cli`) is meant to change the stored values of listed temperature records, but in 62.0.1 it does not change the listed records. See Climate changes.

Example

First lines of `refdata/Osu_1hru/Imsiltmp.tmp` in the SWAT+ repository:

```
Imsiltmp.tmp: Temperature data - file written by SWAT+ editor 2023-03-22 04:22:29.141050
nbyr     tstep       lat       lon      elev
  42         0    35.610   127.290   247.900
1981    1    2.70000   -17.50000
1981    2   -3.00000    -7.50000
...
2023  364        2.3        -4.3
2023  365        0.8        -8.4
```

A daily record from 1 January 1981 to 31 December 2023 at 247.9 m. The record spans 43 calendar years while `nbyr` is 42. The Osu run starts in 2010, so only the 14 years from 2010 are stored and the smaller `nbyr` does no harm here; a run starting in 1981 would need `nbyr` of at least 43.

Related files

- `tmp.cli`: lists the temperature files.
- `weather-sta.cli`: assigns a temperature file to each weather station in its `tmp` column.
- `weather-wgn.cli`: weather generator statistics that fill missing days.
- `*.tem`: the same format under another extension.
- `file.cio`: names `tmp.cli` and the `tmp_path` prefix.
- `codes.bsn` and `parameters.bsn`: `lapse` switch and `tlaps` rate.
- `snow.sno`: snowfall and melt use the daily mean and maximum temperature.

Sources

- Reader: `cli_tmeas.f90`
- Type definition: `type climate_measured_data` in `climate_module.f90`
- Gauge name lookup from `weather-sta.cli`: `cli_staread.f90`, `search.f90`
- Daily use, missing values and range check: `climate_control.f90`, `cli_bounds_check.f90`
- Weather generator: `cli_weatgn.f90`, `cli_tgen.f90`
- Elevation adjustment: `cli_lapse.f90`, applied in `hru_control.f90`
- Path prefix: `in_path_tmp` in `input_file_module.f90`, read in `readcio_read.f90`
- Calibration change `temp`: `cal_conditions.f90`

Column	Fortran variable
`nbyr`	`tmp(i)%nbyr`
`tstep`	`tmp(i)%tstep`
`lat`	`tmp(i)%lat`
`lon`	`tmp(i)%long`
`elev`	`tmp(i)%elev`
tmax	`tmp(i)%ts(day, iyr)`
tmin	`tmp(i)%ts2(day, iyr)`

Purpose

`.tem` is an older extension for temperature data files, used by some data sets and preprocessing tools. SWAT+ opens a temperature file by the name given in `tmp.cli` and does not look at the extension, so a `.tem` file is read in the same way as a `.tmp` file. SWAT+ Editor writes `.tmp`.

The name has to be the same, extension included, in two places:

- a line of `tmp.cli`, which tells SWAT+ to read the file, and
- the `tmp` column of `weather-sta.cli`, which assigns it to a weather station.

If the `tmp` column names a file that is not listed in `tmp.cli`, SWAT+ writes `<name> file not found (tgage)` to `diagnostics.out` and generates the temperatures of that station for the whole run.

Format

The format is that of a .tmp file: a title line, a header line, a gauge line with **nbyr**, **tstep**, **lat**, **lon** and **elev**, and one record per day with year, day of year, maximum and minimum temperature in °C. Values of -97 or less mark missing data.

Example

refdata/Ames_sub1/ames.tem in the SWAT+ repository:

```
AME.Tmp
 NBYR    TSTEP       LAT      LONG      ELEV
   121        0     42.04    -93.89    316
1900    1   -99 -99
1900    2   -99 -99
...
1975    1   4.4 -7.2
1975    2   5   -7.8
...
2020    365 -1.7    -7.8
2020    366 -2.8    -8.9
```

A daily record of 121 years, 1900 to 2020, separated by tabs, which free-format reading accepts. In this data set the **tmp** column of weather-sta.cli names **ames.tem**, while tmp.cli lists **ames.tmp**, which is an empty file. SWAT+ therefore does not find **ames.tem**, never reads it, and generates the temperatures of the Ames run. Listing **ames.tem** in tmp.cli instead of **ames.tmp** makes the run use the measured record.

Related files

- *.tmp: the same format; missing values, elevation adjustment and storage are described there.
- tmp.cli: lists the temperature files to read.
- weather-sta.cli: assigns a temperature file to each weather station.

Sources

- Reader: cli_tmeas.f90
- Type definition: **type climate_measured_data** in climate_module.f90
- Name lookup and the **file not found (tgage)** message: cli_staread.f90, search.f90
- Daily use: climate_control.f90

Purpose

snow.sno is the snow parameter database. Each row is a named set of parameters for snowfall, snowpack temperature, snowmelt and snow cover. An HRU picks a row by name in the **snow** column of hru-data.hru. The file is named on the **hru_parm_db** line of file.cio (default **snow.sno**).

If hru-data.hru gives **null** for an HRU's snow entry, or a name that is not in **snow.sno**, the HRU uses the first row of **snow.sno**. No message is written in either case. A model therefore needs a **snow.sno** file with at least one row: if the file is missing or named **null** in file.cio, every HRU refers to a first row that does not exist.

Format

Free format, whitespace-separated. Line 1 is a title and line 2 is the column header; both are skipped. Each later line is one parameter set with nine values in the order below. The reader does not match columns by header name, so all nine values must be present and in this order.

#	Column	Units	Typical range	Description
1	`name`	-	-	Name of the snow parameter set, matched against the `snow` column of `hru-data.hru`.
2	`fall_tmp`	°C	-5 to 5	Snowfall temperature. On a day when the mean air temperature is at or below `fall_tmp`, all precipitation falls as snow and is added to the snowpack.
3	`melt_tmp`	°C	-5 to 5	Snowmelt base temperature. Melt occurs only on days when the maximum air temperature is above `melt_tmp`.
4	`melt_max`	mm H_2O/°C/day	0 to 10	Melt factor on 21 June (day 172).
5	`melt_min`	mm H_2O/°C/day	0 to 10	Melt factor on 21 December (day 355).
6	`tmp_lag`	-	0.01 to 1	Snowpack temperature lag factor. See Snowpack temperature for how 62.0.1 uses it.
7	`snow_h2o`	mm H_2O	0 to 500	Snow water content at and above which the HRU is fully covered by snow.
8	`cov50`	fraction	0.05 to 0.95	Fraction of `snow_h2o` at which 50% of the HRU is covered by snow.
9	`snow_init`	mm H_2O	0 and up	Snow water content at the start of the simulation.

Snow processes

All snow calculations run daily for each HRU, after canopy interception.

Snowfall

If the mean air temperature $\bar{T}$ is at or below `fall_tmp`, the day's precipitation (after canopy interception) is added to the snowpack and no rain reaches the ground. In sub-daily runs the rainfall for every time step of that day is set to zero.

Snowpack temperature

The intended snowpack temperature keeps part of the previous day's value:

$$T_{snow,d} = T_{snow,d-1}\,(1-\ell) + \bar{T}_d\,\ell$$

where ℓ is `tmp_lag`. Values near 1 give the current day's air temperature most of the weight. Values near 0 make the snowpack respond slowly.

In 62.0.1 the previous day's snowpack temperature is reset to zero at the start of each day, so the calculation reduces to

$$T_{snow,d} = \ell\,\bar{T}_d$$

and `tmp_lag` acts as a scaling factor on the day's mean air temperature. It has no lag effect.

Melt factor

The melt factor b_{mlt} varies through the year between `melt_min` and `melt_max`, which accounts for the effect of snowpack density on melt:

$$b_{mlt} = \frac{\text{melt_max} + \text{melt_min}}{2} + \frac{\text{melt_max} - \text{melt_min}}{2} \sin\left(\frac{2\pi}{365}(d - 81)\right)$$

where d is the day of the year. The factor equals `melt_max` on day 172 and `melt_min` on day 355 for every HRU. SWAT+ 62.0.1 does not swap the two for the Southern Hemisphere, so a model there should give the winter (June) factor in `melt_max` and the summer (December) factor in `melt_min`.

In rural areas the melt factor ranges from 1.4 to 6.9 mm H_2O/°C/day (Huber and Dickinson, 1988). Urban values tend to the high end because traffic compresses the snowpack: urban studies in Sweden reported 3.0 to 8.0 mm H_2O/°C/day (Bengston, 1981; Westerstrom, 1981), and snow on asphalt gave 1.7 to 6.5 mm H_2O/°C/day (Westerstrom, 1984).

Snow cover

Drifting, shading and topography mean that snow rarely covers an HRU evenly. SWAT+ describes the covered fraction with an areal depletion curve. With $r = SNO/\text{snow_h2o}$, the ratio of the current snow water content to `snow_h2o`,

$$sno_{cov} = \frac{r}{r + \exp(c_1 - c_2\, r)} \quad \text{for } r < 1, \qquad sno_{cov} = 1 \quad \text{for } r \geq 1$$

The shape coefficients c_1 and c_2 are set once at the start of the run so that the curve passes through 50% cover at $r = \text{cov50}$ and 95% cover at $r = 0.95$. `cov50` must therefore lie between 0 and 0.95.

The depletion curve affects melt only while the snow water content is between 0 and `snow_h2o`. A small `snow_h2o` makes its effect on melt minor; a large one makes it more important.

Snowmelt

When the maximum air temperature T_{max} is above `melt_tmp` and there is snow on the ground, melt is

$$SNO_{mlt} = b_{mlt}\, sno_{cov} \left(\frac{T_{snow} + T_{max}}{2} - \text{melt_tmp}\right)$$

limited to between zero and the snow water content. Melt water is added to the day's water input at the surface, together with any rain. In sub-daily runs it is spread evenly over the day's time steps.

Calibration and decision tables

Five of the parameters can be changed during calibration through `calibration.cal`, using these parameter names: `snofall_tmp` (`fall_tmp`), `snomelt_tmp` (`melt_tmp`), `snomelt_max` (`melt_max`), `snomelt_min` (`melt_min`) and `snomelt_lag` (`tmp_lag`). `snow_h2o`, `cov50` and `snow_init` have no calibration parameter. See parameter change files.

A decision table action `snow_change` switches an HRU to another `snow.sno` row during the run and logs the change to `lu_change_out.txt`. The snow cover curve is not recalculated after the switch, so it keeps the shape from the HRU's original `cov50`.

Example

`refdata/Ames_sub1/snow.sno` in the SWAT+ repository:

```
snow.sno: written by SWAT+ editor v2.3.3 on 2023-10-25 08:58 for SWAT+ rev.60.5.7
name                  fall_tmp      melt_tmp      melt_max      melt_min       tmp_lag      snow_h2o
snow001                1.00000       0.50000       4.50000       4.50000       1.00000       1.00000
```

One parameter set named `snow001`. `melt_max` and `melt_min` are both 4.5, so the melt factor is constant through the year. With `tmp_lag` at 1.0 the snowpack temperature equals the day's mean air temperature, and with `snow_h2o` at 1 mm the HRU is fully covered by any snowpack of 1 mm or more.

Related files

- `hru-data.hru`: selects a `snow.sno` row by name in its `snow` column; `null` or an unknown name selects the first row.
- `file.cio`: names the snow file.
- `tmp.cli` and `pcp.cli`: supply the air temperatures and precipitation the snow routine uses.

Sources

- Reader: `snowdb_read.f90`
- Type definition: `type snow_parameters` in `hru_module.f90`
- Name lookup from `hru-data.hru`: `hru_read.f90`
- HRU setup and depletion curve: `topohyd_init.f90`, `ascrv.f90`
- Snowfall and melt: `sq_snom.f90`, called from `hru_control.f90`
- Decision table action `snow_change`: `actions.f90`
- Calibration parameters: `cal_parm_select.f90`
- File name slot: `in_parmdb%snow` in `input_file_module.f90`

Column	Fortran variable
`name`	`snodb%name`
`fall_tmp`	`snodb%falltmp`
`melt_tmp`	`snodb%melttmp`
`melt_max`	`snodb%meltmx`
`melt_min`	`snodb%meltmn`
`tmp_lag`	`snodb%timp`
`snow_h2o`	`snodb%covmx`
`cov50`	`snodb%cov50`
`snow_init`	`snodb%init_mm`